Experiment No 5

Title:- Program a login webpage in JSP that makes validation through Database using JDBC and call the servlet for various operations.

Prerequisites

- Apache Tomcat server
- MySQL database (users table)
- JDBC MySQL connector (JAR file in lib)

Steps for Implementation:

Step 1: Install MySQL, go to https://dev.mysql.com/downloads/installer/

And download mysql-installer-web-community-8.0.42.0

Double click and it will get install.

Step 2: To open MySQL, open command prompt and go to folder

C:\Program Files\MySQL\MySQL Server 8.0\bin

and give following command

mysql -u root -p

following screen will appear:

```
C:\Program Files\MySQL\MySQL Server 8.0\bin>mysql -u root -p
Enter password: ****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 9
Server version: 8.0.42 MySQL Community Server - GPL

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> |
```

Step 2: **Create MySQL Database Table**

```
CREATE DATABASE login_db;

USE login_db;


CREATE TABLE users (

    id INT PRIMARY KEY AUTO_INCREMENT,

    username VARCHAR(50) NOT NULL,

    password VARCHAR(50) NOT NULL

);


INSERT INTO users (username, password) VALUES ('admin', 'admin123');
```

The output is as follows:

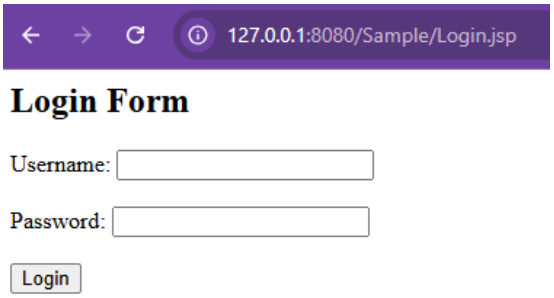
```
mysql> CREATE DATABASE login_db;
Query OK, 1 row affected (0.07 sec)

mysql> USE login_db;
Database changed
mysql> CREATE TABLE users (
    ->     id INT PRIMARY KEY AUTO_INCREMENT,
    ->     username VARCHAR(50) NOT NULL,
    ->     password VARCHAR(50) NOT NULL
    -> );
Query OK, 0 rows affected (0.06 sec)
```

```
mysql> INSERT INTO users (username, password) VALUES ('admin', 'admin123');
Query OK, 1 row affected (0.02 sec)

mysql> |
```

Step 2: Write “Login.jsp” file and save file in C:\apache-tomcat-9.0.106-windows-x64\apache-tomcat-9.0.106\webapps\Sample folder as follows:

```
<%@ page language="java" contentType="text/html; charset=UTF-8" pageEncoding="UTF-8"%>

<html>

<head>

    <title>Login Page</title>

</head>

<body>

    <h2>Login Form</h2>

    <form action="LoginServlet" method="post">

        Username: <input type="text" name="username" required /><br><br>

        Password: <input type="password" name="password" required /><br><br>

        <input type="submit" value="Login" />

    </form>

</body>

</html>
```

Step 2: Open browser and type URL as http://127.0.0.1:8080/Sample/Login.jsp



Step 3: Servlet: LoginServlet.java

Save the file in Tomcat/webapps/Sample/WEB-INF/servlet folder

```
import java.io.IOException;
```

```
import java.sql.*;

import javax.servlet.*;

import javax.servlet.annotation.WebServlet;

import javax.servlet.http.*;


@WebServlet("/LoginServlet")

public class LoginServlet extends HttpServlet {

    private static final String DB_URL = "jdbc:mysql://localhost:3306/login_db";

    private static final String DB_USER = "root";

    private static final String DB_PASSWORD = "your_password"; // Replace with your MySQL
password


    protected void doPost(HttpServletRequest request, HttpServletResponse response)

            throws ServletException, IOException {


        String uname = request.getParameter("username");

        String pwd = request.getParameter("password");


        try {

            Class.forName("com.mysql.cj.jdbc.Driver");

            Connection conn = DriverManager.getConnection(DB_URL, DB_USER,
DB_PASSWORD);

            PreparedStatement ps = conn.prepareStatement("SELECT * FROM users WHERE
username=? AND password=?");

            ps.setString(1, uname);

            ps.setString(2, pwd);
```

```
            ResultSet rs = ps.executeQuery();

            if (rs.next()) {
                request.setAttribute("username", uname);
                RequestDispatcher rd = request.getRequestDispatcher("welcome.jsp");
                rd.forward(request, response);
            } else {
                RequestDispatcher rd = request.getRequestDispatcher("error.jsp");
                rd.forward(request, response);
            }

            rs.close();
            ps.close();
            conn.close();
        } catch (Exception e) {
            throw new ServletException("Login Failed", e);
        }
    }
}
```

Compile the file as follows:

And save LoginServlet.class file in Tomcat\webapps\Sample\WEB-INF\classes folder.

Step 4: Welcome Page: welcome.jsp, Save file in Tomcat\webapps\Sample folder

```
<%@ page language="java" contentType="text/html; charset=UTF-8" pageEncoding="UTF-8"%>

<html>

<head><title>Welcome</title></head>

<body>

    <h2>Welcome, <%= request.getAttribute("username") %>!</h2>

    <p>You have successfully logged in.</p>

</body>

</html>
```

Step 5: Error Page: error.jsp, Save file in Tomcat\webapps\Sample folder

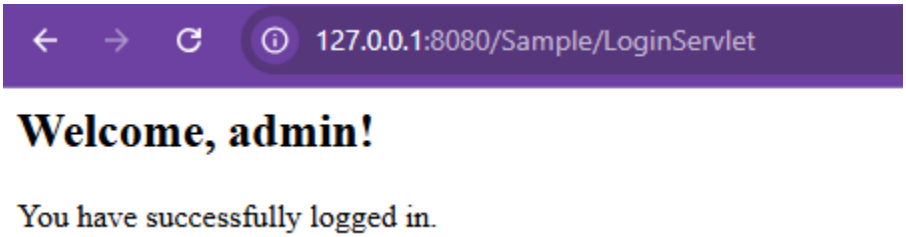
```
<%@ page language="java" contentType="text/html; charset=UTF-8" pageEncoding="UTF-8"%>

<html>

<head><title>Login Failed</title></head>

<body>

    <h2>Login Failed</h2>

    <p>Invalid username or password. Try again.</p>

    <a href="login.jsp">Back to Login</a>

</body>

</html>
```

Step 5: Edit web.xml for Web Deployment as follows:

```
<web-app xmlns="http://xmlns.jcp.org/xml/ns/javaee" version="3.1">

    <servlet>
```

```
        <servlet-name>LoginServlet</servlet-name>
        <servlet-class>com.example.LoginServlet</servlet-class>
    </servlet>
    <servlet-mapping>
        <servlet-name>LoginServlet</servlet-name>
        <url-pattern>/LoginServlet</url-pattern>
    </servlet-mapping>
</web-app>
```

Step 6: Restart the Tomcat server by clicking shutdown.bat file and again click startup.bat file

1. Visit http://localhost:8080/Sample/Login.jsp
2. Try:
    - Username: admin
    - Password: admin123

The output is shown as follows: